

Advanced Optimization — Lecture Plan

Fall 2026, Sogang University, Dept. of Electronic Engineering Prof. Hongseok Kim

Text: Nocedal & Wright, *Numerical Optimization*, 2nd ed. Tue/Thu. Revised Oct 9, 2026, v3 (student presentations Nov 10–17; Ch. 10 if time permits).

Scope. The course covers *unconstrained* optimization in depth (Ch. 3–7) and the *theory* of constrained optimization (Ch. 12). It closes with a two-lecture reading guide to primal-dual interior-point methods (Ch. 19) — the algorithm inside IPOPT and KNITRO — and a final project in which you run IPOPT on a small AC OPF and explain every line of its log using Ch. 3, 12 and 19. Constrained *algorithms* (Ch. 15–18) are outside the course; the last section says when you would need them.

Wk	Date	Topic	Reading / HW
1–4	Sep 1–22	Ch. 1–2; Ch. 3 line search (Wolfe, Zoutendijk, Newton/quasi-Newton rates, Hessian modification); Ch. 4 framework, Algorithm 4.1, Theorem 4.1 — <i>done</i>	HW3 out Sep 22
4	Sep 24	<i>No class (Chuseok)</i>	
5	Sep 29	Ch. 4: Cauchy point, fraction of Cauchy decrease, dogleg, 2D subspace minimization; Steihaug-CG named only — <i>done</i>	§4.1
	Oct 1	Ch. 4: global convergence (Thm. 4.5–4.6), local convergence, scaling; §4.3 skipped — <i>done</i>	§4.2, 4.4–4.5; HW4 out
6	Oct 6	Ch. 5: linear CG I — conjugate directions, expanding subspace (Thm. 5.2), Algorithm 5.1 → 5.2 — <i>done</i>	§5.1
	Oct 8	Ch. 5: linear CG II — rate of convergence, eigenvalue clustering, preconditioning; nonlinear CG — <i>done</i>	§5.1–5.2; HW5 out
7	Oct 13	Ch. 6: BFGS — secant equation, why $s^T y > 0$ needs Wolfe, implementation	§6.1
	Oct 15	Ch. 6: SR1, Broyden class, convergence results (stated), Dennis–Moré revisited	§6.2–6.4; HW6 out
8	Oct 20	Midterm exam (Ch. 3–6)	
	Oct 22	Buffer / catch-up (midterm week). If Ch. 6 is on schedule: Ch. 10 Gauss–Newton and Levenberg–Marquardt	§10.1–10.3 (if time permits)
9	Oct 27	Ch. 7: inexact Newton, Newton-CG, trust-region Newton-CG	§7.1
	Oct 29	Ch. 7: L-BFGS; what is skipped in §7.3–7.4 and why	§7.2; HW7 out
10	Nov 3	Ch. 12: examples, tangent cone, LICQ, first-order necessary conditions (statement)	§12.1–12.2
	Nov 5	Ch. 12: proof of KKT via Farkas’ lemma; constraint qualifications	§12.3, 12.6
11	Nov 10	Student presentation 1: Shampoo (instructor traveling; recorded)	see below
	Nov 12	Student presentation 2: SOAP (instructor traveling; recorded)	see below
12	Nov 17	Student presentation 3: Muon (instructor traveling; recorded)	see below
	Nov 19	Ch. 12: second-order conditions, critical cone	§12.5
13	Nov 24	Ch. 12: degeneracy, duality, multipliers as sensitivities; KKT of a small AC OPF, multipliers as prices	§12.7–12.9, notes; HW12 out
	Nov 26	Ch. 19 reading guide I: barrier problem, primal-dual Newton system, central path, inertia (§3.4 again)	§19.1–19.3
14	Dec 1	Ch. 19 reading guide II: merit function / filter line search; an IPOPT log, line by line	§19.4–19.6; Wächter–Biegler (2006)
	Dec 3	Final project workshop (IPOPT via JuMP / Pyomo / CasADi)	
15	Dec 8, 10	Final project presentations	
16	Dec 15, 17	Final exam week — no lecture; project report due	

Assumed: 16-week semester from Sep 1, midterm week 8 (Oct 20–22), final week 16 (Dec 15–17). Adjust if the official calendar differs. Changes from the Sep 22 version: the instructor is traveling Nov 9–18, so Nov 10, 12 and 17 are student presentations; Ch. 12 continues on Nov 19–24, and Ch. 19 and the project workshop move later by one session. The two buffer sessions (Nov 19, Dec 3) are used for this.

Student presentations (Nov 10, 12, 17). Six students, three groups of two, one optimizer per session:

- **Shampoo** — Gupta, Koren & Singer (2018).
- **SOAP** — Vyas et al. (2024).
- **Muon** — Jordan et al. (2024).

Each group studies its method together and presents it in the language of this course: which matrix plays the role of the preconditioner or of B_k , how it is updated, what one step costs, and how it relates to preconditioning (Ch. 5) and quasi-Newton updating (Ch. 6–7). The session is run by the students and recorded; the group uploads the recording.

Homework thread. One problem set per chapter, one week each. Code is expected to be written with AI assistance; grading is on the *observations*, not the code. The line search from HW3 is reused in HW5–HW7.

- **HW3** — backtracking (Alg. 3.1) and strong-Wolfe (Alg. 3.5–3.6) line searches; steepest descent and Newton on Rosenbrock; Alg. 3.3 on an indefinite Hessian; Zoutendijk annotated; Ex. 3.5.
- **HW4** — Algorithm 4.1 with dogleg on Rosenbrock; $\log \Delta_k, \rho_k$; compare iteration counts with HW3.
- **HW5** — linear CG on an SPD matrix with clustered vs. spread eigenvalues ($n = 100$): iteration count follows the clusters; FR and PR^+ with the HW3 line search, $c_2 = 0.1$.
- **HW6** — BFGS with the HW3 line search on Rosenbrock; $\log \|B_k - \nabla^2 f_k\|$ against $\|(B_k - \nabla^2 f_k)p_k\|/\|p_k\|$ (Dennis–Moré in action).
- **HW7** — L-BFGS with $m \in \{3, 5, 20\}$ on an $n = 10^4$ problem; compare with Newton-CG.
- **HW12** — KKT conditions by hand for a 2- or 3-bus DC OPF; identify what each multiplier means.

Final project. Solve a small AC OPF (pglib-opf `case5` or `case14`) with IPOPT. Annotate the solver log: barrier parameter μ and its decrease (§19.3), primal/dual infeasibility and complementarity (Ch. 12), **inertia correction** (§3.4), step lengths α and the merit/filter decision (§19.4). Report: 4 pages, plus a 10-minute presentation.

If time permits. Ch. 10 (nonlinear least squares: Gauss–Newton, Levenberg–Marquardt). It needs only Ch. 4 and Ch. 6: Gauss–Newton chooses $B_k = J_k^T J_k$, and Levenberg–Marquardt is the trust-region method of Ch. 4 applied to this model. It is taught in a free session before Ch. 12 (first choice: Oct 22).

Not covered, and when you would need it.

- Ch. 8 (derivatives, automatic differentiation) — this is what **autograd** does; read §8.2 once.
- Ch. 9 (derivative-free), Ch. 11 (nonlinear equations) — niche for this course.
- Ch. 13 (simplex), Ch. 14 (LP interior point) — Ch. 14’s idea is all that Ch. 19 needs, and it is given in the reading guide.
- Ch. 15–18 (merit functions, QP, penalty/augmented Lagrangian, SQP) — read Ch. 17 if you use ADMM-type methods; Ch. 18 is KNITRO’s other mode.